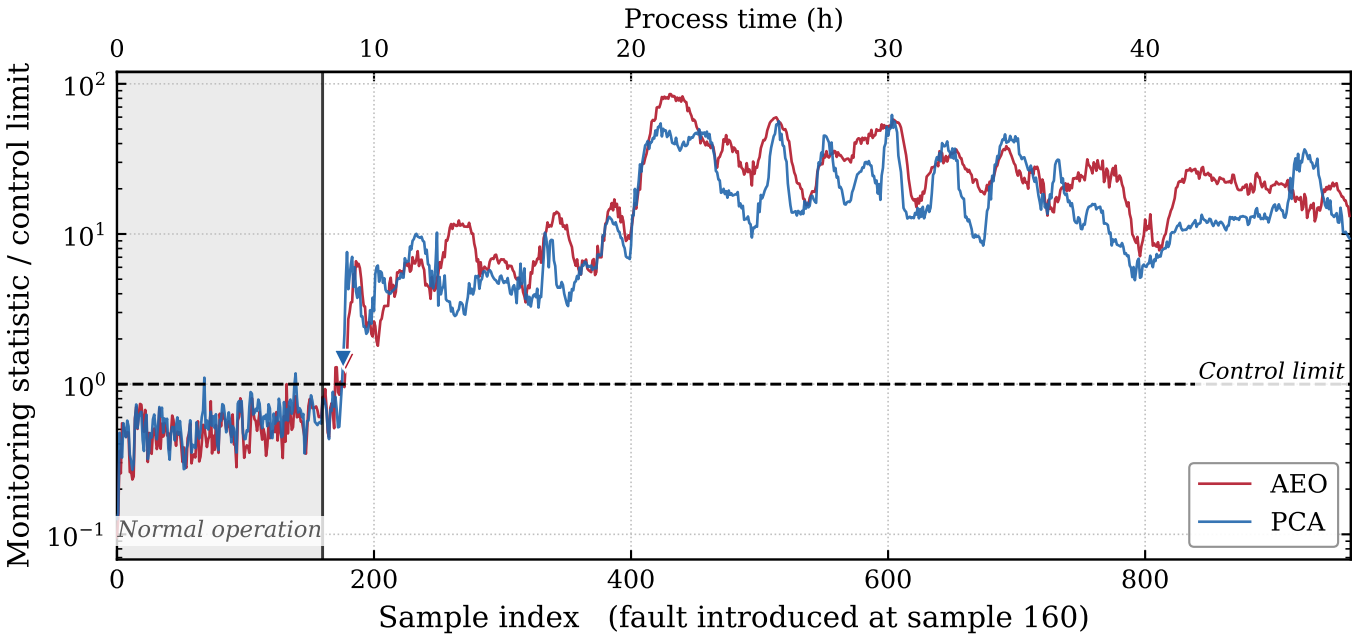
Fault Detection Performance for IDV(8)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	30.6 samples (92 min)	MDR 0.041	detected in 7/7 runs
PCA	delay	24.7 samples (74 min)	MDR 0.034	detected in 7/7 runs

PCA detects 5.9 samples (18 min) earlier; PCA missed-detection rate lower by 18%